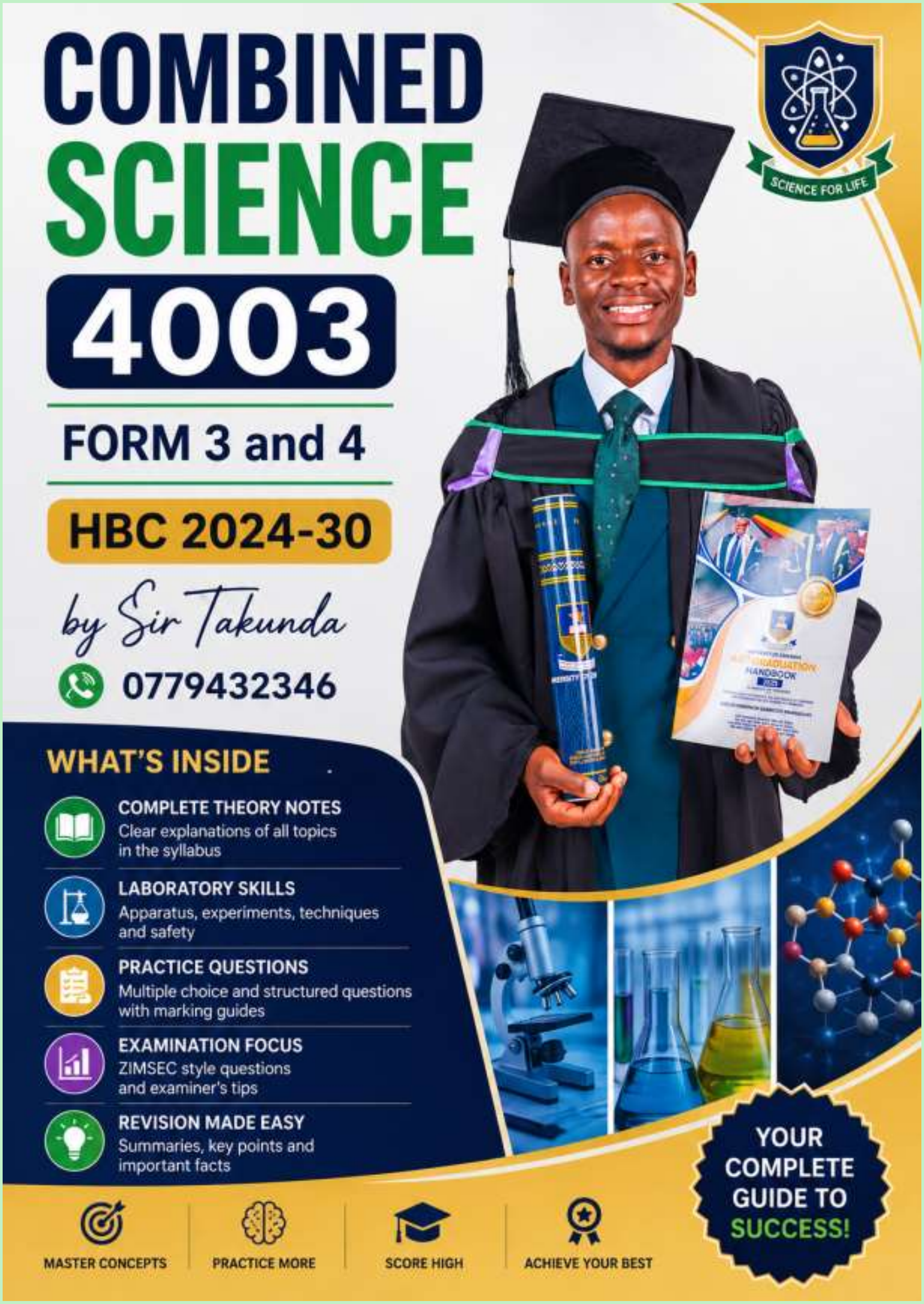

COMBINED
SCIENCE
4003
FORM 3 and 4
HBC 2024-30
by Sir Takunda
0779432346
SCIENCE FOR LIFE
WHAT'S INSIDE
COMPLETE THEORY NOTES
Clear explanations of all topics in the syllabus
LABORATORY SKILLS
Apparatus, experiments, techniques and safety
PRACTICE QUESTIONS
Multiple choice and structured questions with marking guides
EXAMINATION FOCUS
ZIMSEC style questions and examiner's tips
REVISION MADE EASY
Summaries, key points and important facts
MASTER CONCEPTS
PRACTICE MORE
SCORE HIGH
ACHIEVE YOUR BEST
YOUR COMPLETE GUIDE TO SUCCESS!

AUGUST VACATION CLASS ENROLMENT IN PROGRESS

PHYSICAL AND ONLINE CLASSES AVAILABLE

SIR TAKUNDA

0779432346

ENROL TODAY

ZIMSEC COMBINED SCIENCE (4003)

FORM 3 CHEMISTRY

TOPIC: SEPARATION OF SUBSTANCES

Part 1: Distillation

Learning Objectives

By the end of this lesson, learners should be able to:

- Describe the process of simple distillation.
- Explain the principle of simple distillation.
- Identify the apparatus used in simple distillation.
- Draw and label a simple distillation apparatus.
- Describe the procedure of simple distillation.
- State the uses of simple distillation.
- Explain the advantages and limitations of simple distillation.

Introduction

In everyday life, many substances exist as mixtures. A **mixture** is made up of two or more substances that are physically combined and can be separated by physical methods.

Examples include:

- Salt water
- River water
- Air
- Tea
- Milk
- Petrol

Scientists use different methods to separate mixtures depending on the physical properties of their components. One of the most important methods is **distillation**.

Distillation is commonly used in homes, hospitals, industries and laboratories to obtain pure liquids.

What is Distillation?

Definition

Distillation is the process of separating a liquid from a solution by heating it until it boils and then cooling the vapour to obtain the pure liquid.

OR

Distillation is the separation of a solvent from a solution by evaporation followed by condensation.

Example

When salty water is heated,

- water evaporates,
- salt remains in the flask,
- water vapour is cooled,
- pure water is collected.

Thus,

Salt water → Pure water + Salt

Principle of Distillation

Distillation works because different substances have different **boiling points**.

The substance with the **lower boiling point** boils first.

When heated,

- liquid changes into vapour (evaporation)
- vapour moves into the condenser
- cold water cools the vapour
- vapour changes back into liquid (condensation)

The condensed liquid collected is called the **distillate**.

Key Terms

Mixture

A combination of two or more substances that are not chemically combined.

Solution

A mixture formed when a solute dissolves in a solvent.

Solvent

The substance that dissolves another substance.

Example:
Water

Solute

The substance dissolved in a solvent.

Example:
Salt

Vapour

The gaseous form of a liquid.

Condensation

The process by which vapour changes into liquid.

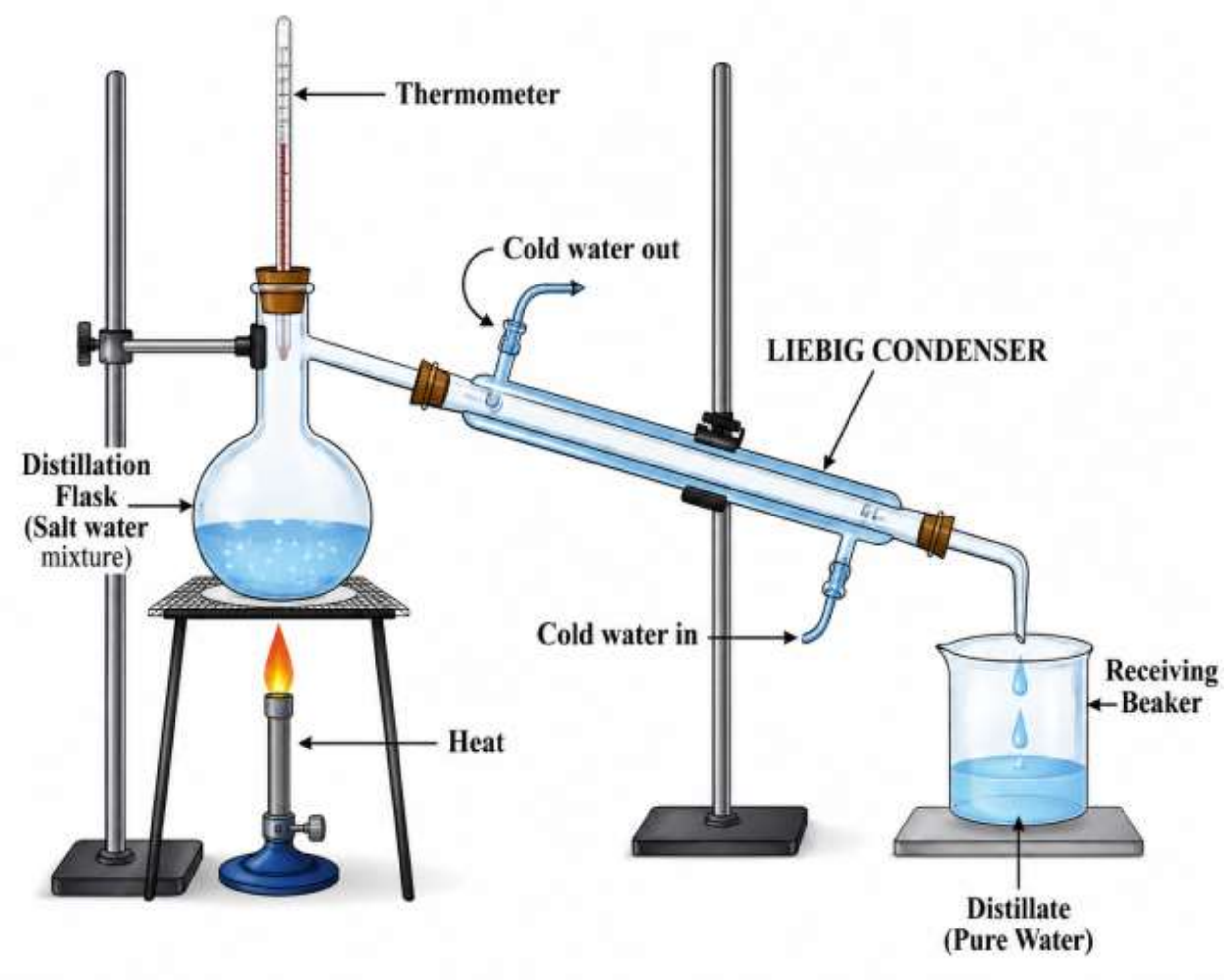
Distillate

The pure liquid collected after condensation.

Apparatus Used

- Distillation flask
- Thermometer
- Liebig condenser
- Beaker/receiver
- Heat source (Bunsen burner or spirit lamp)
- Retort stand
- Rubber tubing

Labelled Diagram of a Simple Distillation Apparatus



Explanation of the Diagram

Distillation Flask

Contains the mixture to be separated.

Thermometer

Measures the temperature of the vapour.

Condenser

Cools vapour into liquid.

Cold Water Inlet

Cold water enters from the lower end.

Cold Water Outlet

Warm water leaves from the upper end.

Receiver

Collects the purified liquid.

Why Does Cold Water Enter from the Bottom?

This is a favourite examination question.

Cold water enters from the bottom so that the condenser fills completely with water.

This provides efficient cooling and ensures maximum condensation.

Procedure

Suppose we want to obtain pure water from salt solution.

Step 1

Pour salt solution into the distillation flask.

Step 2

Assemble the apparatus correctly.

Step 3

Allow cold water to flow through the condenser.

Step 4

Heat the flask gently.

Step 5

Water reaches its boiling point (100°C).

Step 6

Water changes into vapour.

Step 7

Salt does not evaporate because its boiling point is much higher.

Step 8

Water vapour enters the condenser.

Step 9

Cold water cools the vapour.

Step 10

The vapour condenses into pure water.

Step 11

Pure water is collected in the receiver.

Step 12

Salt remains in the flask.

What Happens During Distillation?

Stage	What happens
Heating	Water boils
Evaporation	Water becomes vapour
Condensation	Vapour cools to liquid
Collection	Pure water collected
Residue	Salt remains behind

Why Does Salt Remain Behind?

Salt has

- a very high boiling point,
- does not evaporate at 100°C,
- therefore remains in the flask.

Everyday Uses of Distillation

1. Producing drinking water from sea water

Sea water is distilled to obtain fresh water.

2. Hospitals

Production of distilled water for medicines.

3. Car batteries

Distilled water is used in lead-acid batteries.

4. Laboratories

Preparation of pure water for experiments.

5. Perfume industry

Extraction of perfumes and essential oils.

6. Food industry

Purification of liquids.

Advantages of Distillation

- Produces pure liquids.
- Simple laboratory method.
- Removes dissolved solids.
- Easy to carry out.
- Widely used in industry.

Limitations

- Uses a lot of heat energy.
- Slow process.
- Cannot separate liquids with similar boiling points.
- Requires special apparatus.

Safety Precautions

- Wear safety goggles.
- Clamp apparatus securely.
- Never heat a closed system.
- Handle hot glass carefully.
- Ensure water flows through the condenser before heating.
- Do not point the apparatus toward anyone.

Examination Tips

Remember the sequence:

Heating → Evaporation → Condensation → Collection

Learners often confuse evaporation and condensation. A simple way to remember is:

- **Evaporation:** Liquid → Gas
- **Condensation:** Gas → Liquid

Common Mistakes

1. Saying salt evaporates with water.
2. Confusing boiling with evaporation.
3. Reversing the direction of cooling water.
4. Forgetting to label the condenser.
5. Calling the collected liquid "condensed water" instead of **distillate**.

Summary

- Distillation separates a solvent from a solution.
- It works because substances have different boiling points.
- Water boils at **100°C** under normal atmospheric pressure.
- The vapour is cooled in a condenser.
- The condensed liquid is called the **distillate**.
- Salt remains in the flask because it does not evaporate at the boiling point of water.
- Distillation is used in laboratories, hospitals, industries, and water purification.

End-of-Section Review Questions

Multiple Choice

1. Distillation separates substances based on differences in:
   A. Colour
   B. Density
   C. Boiling point
   D. Mass
2. The pure liquid collected during distillation is called:
   A. Residue
   B. Solute
   C. Distillate
   D. Solvent
3. Which apparatus cools vapour back into liquid?
   A. Beaker
   B. Condenser
   C. Thermometer
   D. Flask
4. Why does salt remain in the flask during distillation?
   A. It is lighter than water.
   B. It has a higher boiling point than water.
   C. It evaporates first.
   D. It dissolves in vapour.
5. In a Liebig condenser, cold water enters from the:
   A. Top
   B. Middle
   C. Bottom
   D. Side

Structured Questions

1. Define distillation.
2. State the principle of distillation.
3. Explain why water enters the condenser from the bottom.
4. Name four uses of distillation.
5. Draw and label a simple distillation apparatus.
6. Explain why salt remains behind during the distillation of salt water.
7. Describe how pure water is obtained from a salt solution using distillation.

PART 2: FRACTIONAL DISTILLATION

Learning Objectives

By the end of this lesson, learners should be able to:

- Describe the process of fractional distillation.
- Explain the principle of fractional distillation.
- Identify and describe the function of a fractionating column.
- Draw and label a fractional distillation apparatus.
- Explain how ethanol is separated from water by fractional distillation.
- Explain how crude oil is separated into useful fractions.
- Compare simple distillation with fractional distillation.
- State the applications of fractional distillation.

Introduction

In Part 1, we learnt that **simple distillation** separates a solvent from a solution or liquids whose boiling points differ greatly.

However, when two liquids have **boiling points that are close together**, simple distillation does not separate them effectively.

For example:

- Ethanol boils at **78°C**
- Water boils at **100°C**

Their boiling points differ by only **22°C**, so both liquids tend to evaporate together. To separate such mixtures, scientists use **fractional distillation**.

What is Fractional Distillation?

Definition

Fractional distillation is the process of separating a mixture of **miscible liquids** (liquids that mix completely) that have **different but close boiling points**.

Unlike simple distillation, fractional distillation uses a **fractionating column** to improve separation.

Principle of Fractional Distillation

Fractional distillation works because different liquids have different boiling points.

When the mixture is heated:

- The liquid with the **lowest boiling point** evaporates first.
- The vapour rises through the fractionating column.
- Inside the column, repeated condensation and evaporation occur.
- The more volatile liquid (lower boiling point) becomes richer in the vapour.
- The less volatile liquid condenses and flows back into the flask.
- Eventually, almost pure vapour of the lower-boiling liquid reaches the condenser and is collected.

Key Terms

Miscible Liquids

Liquids that mix completely to form one uniform solution.

Examples

- Ethanol and water
- Acetone and water

Volatile Liquid

A liquid that evaporates easily because it has a **low boiling point**.

Example:

Ethanol

Less Volatile Liquid

A liquid with a **higher boiling point**.

Example:

Water

The Fractionating Column

The **fractionating column** is the most important part of a fractional distillation apparatus.

It is a vertical tube placed between the distillation flask and the condenser.

The inside is packed with:

- glass beads
- porcelain chips
- metal plates
- wire gauze

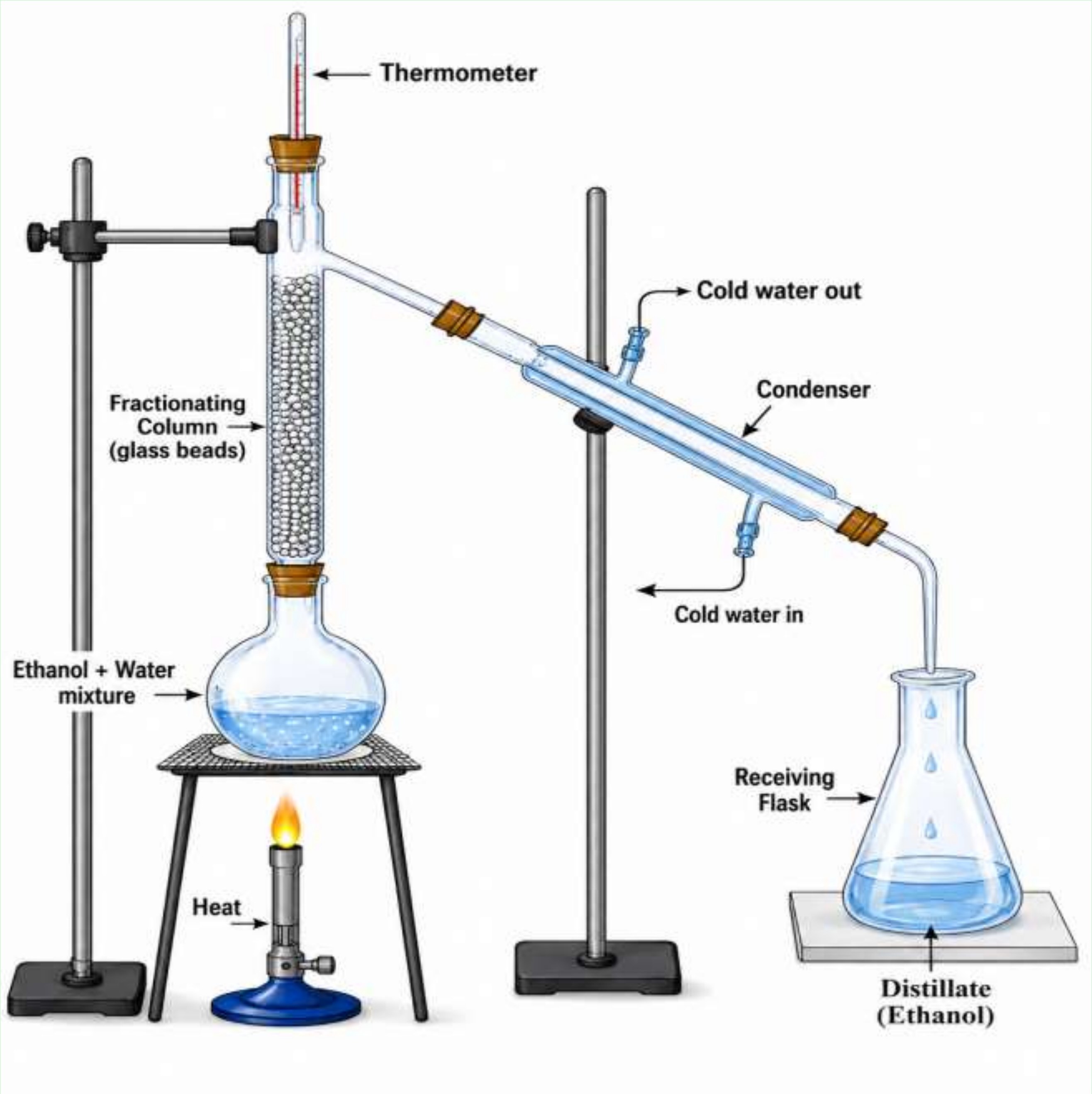
These materials provide a **large surface area** for repeated evaporation and condensation.

Functions of the Fractionating Column

The fractionating column:

- provides a large surface area
- allows repeated condensation
- allows repeated evaporation
- improves separation
- increases the purity of the distillate
- prevents large amounts of water vapour from reaching the condenser

Labelled Diagram of Fractional Distillation



Explanation of the Apparatus

Distillation Flask

Contains the ethanol-water mixture

Fractionating Column

Separates vapours through repeated evaporation and condensation.

Thermometer

Measures the temperature of the vapour entering the condenser.

Liebig Condenser

Cools vapour into liquid.

Receiver

Collects purified ethanol.

How the Fractionating Column Works

Imagine the column as a staircase.

As vapour rises:

- it cools
- condenses
- reheats
- evaporates again

Each evaporation contains **more ethanol** than before because ethanol boils at a lower temperature.

After many repetitions, almost pure ethanol reaches the condenser.

Separation of Ethanol and Water

This is the most important ZIMSEC example.

Step 1

Pour ethanol-water mixture into the flask.

Step 2

Assemble the apparatus.

Step 3

Allow cold water to circulate in the condenser.

Step 4

Heat the mixture gently.

Step 5

At about **78°C**, ethanol begins to boil.

Step 6

Water evaporates much less because its boiling point is **100°C**.

Step 7

Mixed vapours enter the fractionating column.

Step 8

Water vapour condenses inside the column and flows back.

Step 9

Ethanol vapour continues upward.

Step 10

Ethanol enters the condenser.

Step 11

Cold water cools the ethanol vapour.

Step 12

Liquid ethanol is collected.

Step 13

Water remains mainly in the distillation flask.

Why is Ethanol Collected First?

Because ethanol has a **lower boiling point (78°C)** than water **(100°C)**.

Therefore, ethanol evaporates first.

Boiling Points of Common Liquids

Liquid	Boiling Point (°C)
Diethyl ether	35
Ethanol	78
Water	100
Propanol	97
Butanol	118

Fractional Distillation of Crude Oil

Crude oil is a mixture of many hydrocarbons.

Each hydrocarbon has a different boiling point.

In an oil refinery:

- crude oil is heated to about **350–400°C**
- most of it vaporises
- vapours enter a tall **fractionating tower**
- vapours cool as they rise
- each fraction condenses at a different level
- heavier fractions condense lower down
- lighter fractions condense higher up

Main Fractions of Crude Oil

Fraction	Approximate Boiling Range (°C)	Main Uses
Refinery gases	Below 40	Cooking gas (LPG)
Petrol (Gasoline)	40–110	Cars
Naphtha	70–180	Chemical industry
Kerosene	150–250	Jet fuel, lamps
Diesel	250–350	Trucks, buses
Lubricating oil	300–370	Lubricants
Fuel oil	Above 350	Ships, factories
Bitumen	Residue	Road surfacing, roofing

Industrial Uses of Fractional Distillation

1. Separation of ethanol from water.
2. Refining crude oil.
3. Production of petrol.
4. Production of diesel.
5. Production of kerosene.
6. Manufacture of lubricating oils.
7. Separation of liquid air into oxygen, nitrogen and argon (advanced application).

Differences Between Simple and Fractional Distillation

Feature	Simple Distillation	Fractional Distillation
Mixture separated	Liquid from solution or liquids with widely different boiling points	Miscible liquids with close boiling points
Fractionating column	Not used	Used
Separation	One evaporation and one condensation	Repeated evaporation and condensation
Purity	Lower	Higher
Example	Salt water	Ethanol and water

Advantages of Fractional Distillation

- Produces purer liquids.
- Separates liquids with close boiling points.
- Widely used in industries.
- Efficient when separating several liquids.

Limitations

- More expensive than simple distillation.
- Takes longer.
- Requires more apparatus.
- Uses more energy.

Examination Tips

Remember:

- **Simple distillation** → Solution or liquids with **widely different** boiling points.
- **Fractional distillation** → **Miscible liquids** with **close** boiling points.
- The **fractionating column** is the key feature that distinguishes fractional distillation from simple distillation.

Common Learner Mistakes

1. Confusing simple and fractional distillation.
2. Stating that water is collected before ethanol.
3. Omitting the fractionating column in diagrams.
4. Not explaining repeated evaporation and condensation.
5. Assuming both liquids boil at exactly the same

Summary

- Fractional distillation separates **miscible liquids** with **close boiling points**.

- It uses a **fractionating column** to achieve repeated evaporation and condensation.
- Ethanol (boiling point **78°C**) is collected before water (boiling point **100°C**).
- Crude oil is separated into useful fractions using fractional distillation.
- Fractional distillation produces a purer product than simple distillation.

Examination-Style Questions

Section A: Multiple Choice

1. Fractional distillation is used to separate:
   A. Sand and water
   B. Salt and water
   C. Ethanol and water
   D. Iron and sulphur
2. The purpose of the fractionating column is to:
   A. Heat the mixture
   B. Cool the vapour
   C. Allow repeated evaporation and condensation
   D. Collect the distillate
3. Which liquid is collected first from an ethanol-water mixture?
   A. Water
   B. Ethanol
   C. Both together
   D. Salt
4. Which property makes fractional distillation possible?
   A. Colour
   B. Density
   C. Solubility
   D. Boiling point
5. Which fraction is mainly used as fuel for cars?
   A. Bitumen

B. Petrol
C. Lubricating oil
D. Fuel oil

Section B: Structured Questions

1. Define fractional distillation. (2 marks)
2. Explain why a fractionating column is necessary when separating ethanol from water. (4 marks)
3. State four functions of the fractionating column. (4 marks)
4. Describe how ethanol is separated from water by fractional distillation. (6 marks)
5. Explain how crude oil is separated into different fractions. (6 marks)
6. Give three industrial uses of fractional distillation. (3 marks)
7. Compare simple distillation with fractional distillation. (6 marks)

Marking Guide

Question 1 (2 marks)

- Fractional distillation is the separation of **miscible liquids** (1)
- with **different but close boiling points** using a **fractionating column** (1)

Question 2 (4 marks)

Award 1 mark for each correct point:

- Provides a large surface area.
- Causes repeated evaporation.
- Causes repeated condensation.
- Improves purity by allowing less volatile liquid to return to the flask.

Question 3 (4 marks)

Any four correct functions:

- Large surface area.
- Repeated condensation.
- Repeated evaporation.
- Improves separation.
- Increases purity.
- Returns less volatile liquid to the flask.

Question 4 (6 marks)

Award 1 mark for each correct stage:

- Mixture placed in flask.
- Heated gently.
- Ethanol boils first.
- Vapours rise through the fractionating column.
- Ethanol vapour reaches the condenser.
- Condensed ethanol collected in the receiver.

FORM 4 CHEMISTRY

TOPIC: SEPARATION OF SUBSTANCES

PAPER CHROMATOGRAPHY

Learning Objectives

By the end of this lesson, learners should be able to:

- Define paper chromatography.
- Explain the principle of paper chromatography.
- Identify the parts of a paper chromatogram.
- Describe the process of paper chromatography.
- Explain how dyes and plant extracts are separated.
- State the applications of paper chromatography.
- Interpret simple chromatograms.
- Carry out a chromatography experiment safely.

Introduction

Many substances that appear to be a single colour are actually mixtures of several coloured substances called **pigments**.

For example,

- black ink contains several coloured dyes,
- green leaves contain several pigments such as chlorophyll and carotene.

Scientists use **paper chromatography** to separate these coloured substances.

Paper chromatography is one of the simplest and most useful separation techniques used in schools, laboratories, hospitals and industries.

What is Chromatography?

The word **chromatography** comes from two Greek words:

- *Chroma* = colour
- *Graphy* = writing

Originally, chromatography was used to separate coloured substances, but today it is used to separate both coloured and colourless substances

Definition

Paper chromatography is a method of separating the components of a mixture by allowing a solvent to carry them at different speeds across chromatography paper.

OR

Paper chromatography is the separation of dissolved substances according to their different solubilities and attractions to the chromatography paper.

Principle of Paper Chromatography

Paper chromatography works because different substances:

- dissolve differently in the solvent,
- have different attractions to the chromatography paper.

Some substances:

- dissolve more easily,
- move faster,
- travel further.

Others:

- dissolve less easily,
- stick more strongly to the paper,
- move more slowly.

As a result, the mixture separates into different coloured spots.

Important Terms

Chromatogram

The final paper showing the separated spots after chromatography.

Stationary Phase

The substance that does not move.

In paper chromatography,

the filter paper is the stationary phase.

Mobile Phase

The substance that moves.

In paper chromatography,

the solvent is the mobile phase.

Examples include:

- water
- ethanol
- acetone
- benzene
- toluene

Solvent

The liquid that carries the dissolved substances up the paper.

Solvent Front

The highest point reached by the solvent before the experiment is stopped.

The solvent front must always be marked immediately using a pencil.

Baseline (Origin)

A pencil line drawn near the bottom of the paper where the sample is placed.

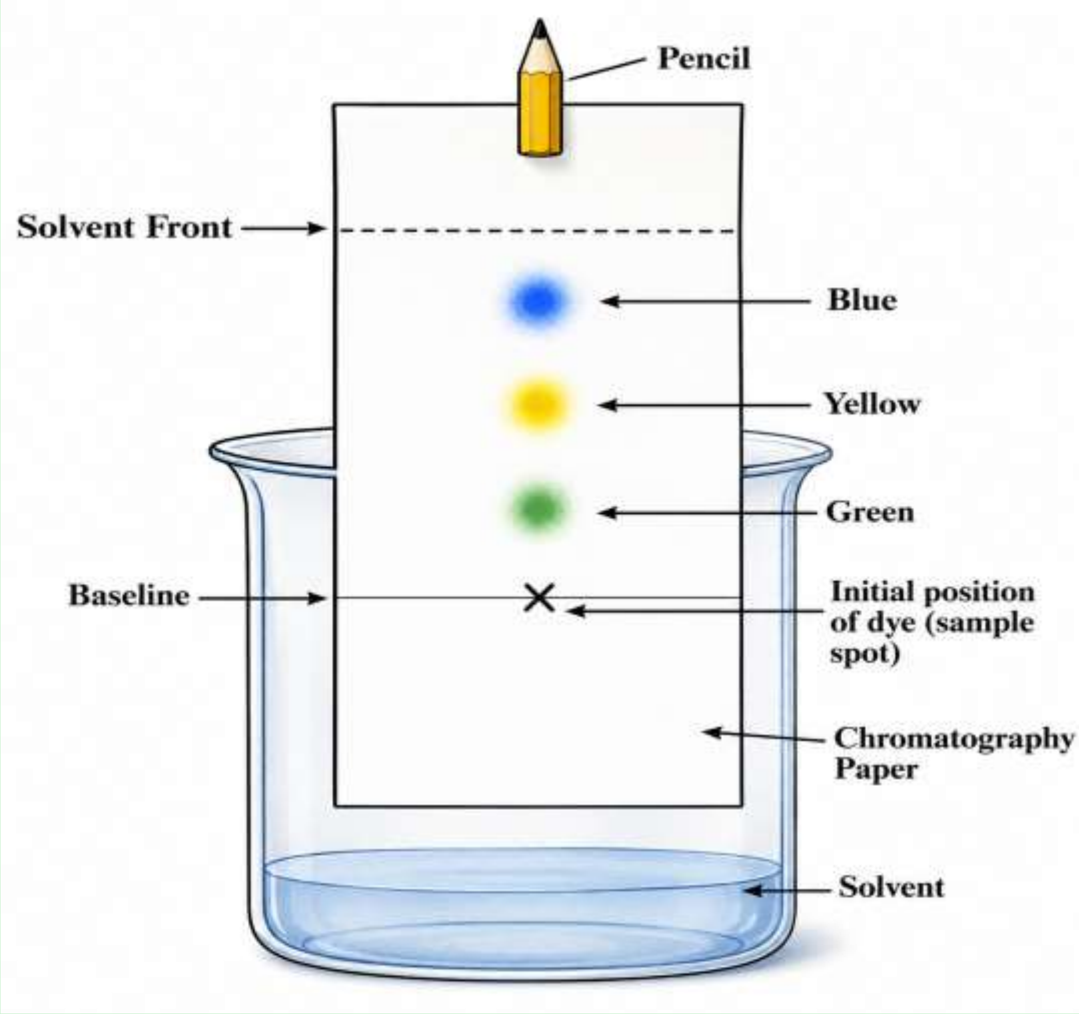
Sample Spot

The small drop of ink or plant extract placed on the baseline.

Apparatus Required

- Filter paper (chromatography paper)
- Beaker
- Solvent
- Pencil
- Ruler
- Capillary tube or dropper
- Mixture of dyes
- Plant extract
- Cover or watch glass

Labelled Diagram of Paper Chromatography



Explanation of the Diagram

Solvent Front

Shows how far the solvent travelled.

Baseline

The starting line drawn in pencil.

Sample Spot

The original mixture placed on the baseline.

Separated Spots

Different dyes separated from the original mixture.

Why is Pencil Used?

This is one of the most common examination questions.

A **pencil** is used because graphite does not dissolve in the solvent.

If ink were used, it would dissolve and interfere with the results.

Why Must the Sample Spot Be Above the Solvent?

The sample must not be submerged.

If it touches the solvent:

- the sample dissolves into the solvent,
- it washes away,
- separation cannot occur properly.

Procedure

Separating Black Ink

Step 1

Draw a pencil line about 2 cm from the bottom of the paper.

Step 2

Place a small spot of black ink on the line.

Step 3

Pour a small amount of solvent into the beaker

Step 4

Suspend the paper so that only the bottom edge touches the solvent.

Step 5

Ensure the ink spot remains above the solvent.

Step 6

Cover the beaker.

Step 7

Allow the solvent to rise up the paper.

Step 8

Different dyes move at different speeds.

Step 9

Remove the paper before the solvent reaches the top.

Step 10

Mark the solvent front immediately.

Step 11

Allow the paper to dry.

Step 12

Observe the separated coloured spots.

How Separation Occurs

The solvent moves upwards by **capillary action**.

As it moves,

- it dissolves the dyes,
- carries them upwards.

Each dye moves a different distance depending on:

- its solubility,
- its attraction to the paper.

Capillary Action

Capillary action is the movement of a liquid through narrow spaces without the help of external forces.

In chromatography, capillary action causes the solvent to move up the paper.

Separation of Plant Pigments

Chromatography is also used to separate pigments found in leaves.

Green leaves contain several pigments, including:

- Chlorophyll a (blue-green)
- Chlorophyll b (yellow-green)
- Xanthophyll (yellow)
- Carotene (orange)

Although a leaf appears green, chromatography shows that it contains several pigments.

Applications of Paper Chromatography

Paper chromatography is used in:

Medicine

- Testing blood samples.
- Detecting drugs.

Food Industry

- Testing food colourings.
- Detecting food additives.

Police Forensics

- Identifying inks used in forged documents.
- Analysing evidence from crime scenes.

Agriculture

- Studying plant pigments.

Environmental Science

- Detecting pollutants in water.

Pharmaceutical Industry

- Checking the purity of medicines.

Thin Layer Chromatography (TLC)

Thin layer chromatography works on the same principle as paper chromatography.

Instead of paper, it uses a thin layer of silica gel or alumina coated on a glass or plastic plate.

TLC is:

- faster,
- more accurate,
- produces better separation than paper chromatography.

Solvents Used in Chromatography

Common solvents include:

- Water
- Ethanol
- Acetone
- Benzene
- Toluene

Teacher's Note: Benzene is effective but highly toxic and carcinogenic. In school laboratories, safer alternatives such as ethanol or suitable chromatography solvents are preferred wherever possible.

Advantages of Paper Chromatography

- Simple to perform.
- Inexpensive.
- Requires little equipment.
- Separates very small amounts of substances.
- Produces accurate results.
- Useful in many scientific fields.

Limitations

- Takes time.
- Some substances cannot be separated.
- Results depend on the solvent used.
- Paper may tear easily.
- Less accurate than advanced laboratory techniques.

Examination Tips

Remember:

- **Stationary phase = Paper**
- **Mobile phase = Solvent**
- **Solvent front = Furthest point reached by the solvent**
- **Baseline = Pencil line**
- **Sample spot = Mixture placed on the baseline**

Common Learner Mistakes

1. Drawing the baseline with ink.
2. Allowing the sample spot to touch the solvent.
3. Forgetting to mark the solvent front.
4. Placing a very large sample spot.
5. Tilting the chromatography paper.
6. Confusing the solvent front with the baseline.

Summary

- Paper chromatography separates mixtures based on differences in solubility and attraction to the paper.
- The paper is the stationary phase.
- The solvent is the mobile phase.
- The solvent moves up the paper by capillary action.
- Different substances travel different distances, producing separate spots.
- Paper chromatography is widely used in medicine, food analysis, forensic science, agriculture, and environmental testing.

Examination-Style Questions

Section A: Multiple Choice

1. Paper chromatography separates substances based mainly on differences in:
   A. Colour
   B. Solubility and attraction to the paper
   C. Density
   D. Melting point
2. Which part of the chromatography setup is the stationary phase?
   A. Solvent
   B. Filter paper
   C. Beaker
   D. Ink

3. Why is the baseline drawn with a pencil?
   A. It is cheaper.
   B. Graphite does not dissolve in the solvent.
   C. It is darker.
   D. It dries quickly.
4. The solvent moves up the paper by:
   A. Diffusion
   B. Osmosis
   C. Capillary action
   D. Condensation
5. Which of the following is **not** an application of paper chromatography?
   A. Testing food colourings
   B. Identifying pigments in leaves
   C. Separating sand from water
   D. Detecting drugs in blood

Section B: Structured Questions

1. Define paper chromatography. (2 marks)
2. State the functions of the:
   - (a) stationary phase (1 mark)
   - (b) mobile phase (1 mark)
3. Explain why the sample spot must be above the solvent level. (2 marks)
4. Describe how you would separate the dyes in black ink using paper chromatography. (6 marks)
5. Name four applications of paper chromatography. (4 marks)
6. Explain why different dyes travel different distances on the chromatography paper. (4 marks)

Marking Guide

Question 1 (2 marks)

Award:

- 1 mark for stating that paper chromatography is a method of separating components of a mixture.
- 1 mark for explaining that separation occurs because substances move at different speeds due to differences in solubility and attraction to the paper.

Question 2 (2 marks)

- (a) Stationary phase: Filter paper (1)
- (b) Mobile phase: Solvent (1)

Question 3 (2 marks)

Award 1 mark each for:

- The sample must not dissolve directly into the solvent.
- The solvent must carry the sample up the paper for separation.

Question 4 (6 marks)

Award 1 mark for each correct stage:

- Draw pencil baseline.
- Place a small sample spot on the baseline.
- Add solvent below the baseline.
- Suspend the paper with the spot above the solvent.
- Allow the solvent to rise and separate the dyes.
- Mark the solvent front and allow the paper to dry.

ZIMSEC COMBINED SCIENCE (4003)

FORM 3 & FORM 4 CHEMISTRY

SEPARATION OF SUBSTANCES

Paper 1 Revision Questions

Part 1 (Questions 1–10)

Time: 10 Minutes
Marks: 10

Instructions

Choose the correct answer from **A, B, C,** and **D**.

Question 1

Which property is mainly used in **simple distillation** to separate substances?

A. Density

B. Colour

C. Boiling point

D. Magnetism

Question 2

During the simple distillation of salt water, the substance collected in the receiving flask is:

A. Salt

B. Pure water

C. Salt solution

D. Steam

Question 3

Why is cold water allowed to enter the condenser from the **bottom**?

A. To make the water flow faster

B. To cool the condenser evenly by filling it completely

C. To increase the boiling point of water

D. To stop evaporation

Question 4

Which apparatus measures the temperature of the vapour during distillation?

A. Beaker

B. Thermometer

C. Condenser

D. Retort stand

Question 5

Fractional distillation is mainly used to separate:

A. Sand and water

B. Salt and water

C. Miscible liquids with close boiling points

D. Iron filings and sulphur

Question 6

Which part of the fractional distillation apparatus improves the separation of liquids?

A. Receiver

B. Condenser

C. Fractionating column

D. Thermometer

Question 7

Ethanol is collected before water during fractional distillation because ethanol:

A. Is heavier than water

B. Has a lower boiling point

C. Is insoluble in water

D. Has a higher density

Question 8

Which of the following is an important use of fractional distillation?

A. Separating sand from gravel

B. Purifying sugar

C. Refining crude oil

D. Separating iron from sulphur

Question 9

Paper chromatography separates substances because they have different:

A. Colours only

B. Densities

C. Solubilities and attractions to the paper

D. Melting points

Question 10

In paper chromatography, the **stationary phase** is the:

A. Solvent

B. Filter paper

C. Ink

D. Dye

ANSWERS AND EXPLANATIONS

Question 1

Correct Answer: C. Boiling point

Explanation

Simple distillation separates substances because they have **different boiling points**. The substance with the lower boiling point evaporates first and is then condensed into a liquid.

Why the others are incorrect

A. Density – Density is used in methods such as separating immiscible liquids with a separating funnel, not in distillation.

B. Colour – Colour does not affect separation by distillation.

D. Magnetism – Magnetism is only useful for separating magnetic materials such as iron.

Examination Tip: *Whenever you see the word* ***distillation****, think* ***boiling point****.*

Question 2

Correct Answer: B. Pure water

Explanation

When salt water is heated, only the water evaporates because it has a much lower boiling point than salt. The water vapour condenses to form **pure water**, which is collected in the receiving flask.

Why the others are incorrect

A. Salt – Salt remains in the distillation flask.

C. Salt solution – The collected liquid is no longer a solution; it is purified water.

D. Steam – Steam is the vapour before condensation. The receiver collects liquid water, not steam.

Examination Tip: *The liquid collected after condensation is called the* ***distillate****.*

Question 3

Correct Answer: B. To cool the condenser evenly by filling it completely

Explanation

Cold water enters from the bottom of the condenser so that the cooling jacket fills completely. This ensures efficient cooling and condensation of the vapour.

Why the others are incorrect

A. Water speed is not the main reason.

C. Cooling water does not change the boiling point.

D. Cooling the condenser does not stop evaporation; it condenses the vapour.

Examination Tip: *Cold water always enters at the **bottom** and leaves at the **top** of a Liebig condenser.*

Question 4

Correct Answer: B. Thermometer

Explanation

The thermometer measures the temperature of the vapour leaving the distillation flask. This helps identify when a particular liquid is boiling.

Why the others are incorrect

A. Beaker – Used to collect the distillate.

C. Condenser – Cools vapour into liquid.

D. Retort stand – Supports the apparatus.

Question 5

Correct Answer: C. Miscible liquids with close boiling points

Explanation

Fractional distillation separates liquids that mix completely and have similar boiling points, such as ethanol and water.

Why the others are incorrect

A. Sand and water – Separated by filtration.

B. Salt and water – Separated by simple distillation or evaporation.

D. Iron filings and sulphur – Separated using a magnet.

Examination Tip: *Simple distillation = one liquid from a solution. Fractional distillation = two or more miscible liquids with close boiling points.*

Question 6

Correct Answer: C. Fractionating column

Explanation

The fractionating column provides a large surface area for repeated evaporation and condensation, improving the separation of liquids.

Why the others are incorrect

A. Receiver – Collects the distillate only.

B. Condenser – Cools vapour but does not improve the separation.

D. Thermometer – Measures temperature.

Question 7

Correct Answer: B. Has a lower boiling point

Explanation

Ethanol boils at approximately **78°C**, while water boils at **100°C**. Ethanol therefore vaporises first and is collected before water.

Why the others are incorrect

A. Being heavier does not determine which liquid boils first.

C. Ethanol is completely miscible with water.

D. Density is not the property used in fractional distillation.

Examination Tip: *Remember the boiling points: Ethanol = **78°C**, Water = **100°C**.*

Question 8

Correct Answer: C. Refining crude oil

Explanation

Oil refineries use fractional distillation to separate crude oil into useful fractions such as petrol, kerosene, diesel and bitumen.

Why the others are incorrect

A. Sand and gravel are separated by sieving.

B. Sugar is purified by crystallisation.

D. Iron and sulphur are separated using a magnet.

Question 9

Correct Answer: C. Solubilities and attractions to the paper

Explanation

Paper chromatography works because different substances dissolve to different extents in the solvent and have different attractions to the chromatography paper. These differences cause them to move at different speeds.

Why the others are incorrect

A. Colour helps us see the spots but does not cause the separation.

B. Density is not the basis of chromatography.

D. Melting point is unrelated to paper chromatography.

Examination Tip: *Chromatography depends on two ideas:* ***solubility*** *and* ***attraction to the stationary phase****.*

Question 10

Correct Answer: B. Filter paper

Explanation

The filter paper does not move during the experiment, so it is the **stationary phase**. The solvent moves up the paper and is called the **mobile phase**.

Why the others are incorrect

A. Solvent – This is the mobile phase.

C. Ink – The sample being separated.

D. Dye – One of the components of the sample.

Examination Tip: *Stationary = Stays still. Mobile = Moves.*

Question 11

Which one of the following mixtures is **best separated by simple distillation**?

A. Ethanol and water

B. Salt and water

C. Sand and iron filings

D. Oil and water

Question 12

Which one of the following remains in the distillation flask after salt solution has been distilled?

A. Water vapour

B. Distilled water

C. Salt crystals

D. Steam

Question 13

The main purpose of a condenser during distillation is to:

A. Heat the mixture

B. Cool vapour into liquid

C. Measure the boiling point

D. Hold the flask firmly

Question 14

During fractional distillation, repeated evaporation and condensation occur mainly inside the:

A. Thermometer

B. Receiver

C. Fractionating column

D. Distillation flask

Question 15

Which statement about ethanol is **correct**?

A. It boils at a higher temperature than water.

B. It boils at about 78°C.

C. It freezes before water during distillation.

D. It has no boiling point.

Question 16

The **solvent front** in paper chromatography is:

A. The point where the sample was placed.

B. The highest point reached by the solvent.

C. The first coloured spot.

D. The bottom of the filter paper

Question 17

Why should the solvent level be below the baseline during chromatography?

A. To increase evaporation.

B. To stop the paper from tearing.

C. To prevent the sample from dissolving directly into the solvent.

D. To reduce the boiling point of the solvent.

Question 18

Which one of the following is **not** an application of paper chromatography?

A. Identifying food colourings

B. Separating pigments in leaves

C. Detecting drugs in blood

D. Separating stones from sand

Question 19

Which process causes the solvent to move up the chromatography paper?

A. Filtration

B. Diffusion

C. Capillary action

D. Sedimentation

Question 20

Which statement correctly compares simple and fractional distillation?

A. Both use a fractionating column.

B. Fractional distillation is used for liquids with similar boiling points.

C. Simple distillation separates crude oil.

D. Fractional distillation is only used in laboratories.

ANSWERS AND EXPLANATIONS

Question 11

Correct Answer: B. Salt and water

Explanation

Simple distillation is used to separate a **solvent from a solution**, such as obtaining pure water from salt solution. Water evaporates and condenses, leaving the salt behind.

Why the others are incorrect

A. Ethanol and water – These liquids have close boiling points and require **fractional distillation**.

C. Sand and iron filings – These are separated using a **magnet**.

D. Oil and water – These are immiscible liquids and are separated using a **separating funnel**.

Exam Tip: Salt solution → **Simple distillation**.

Question 12

Correct Answer: C. Salt crystals

Explanation

Salt has a much higher boiling point than water, so it does not evaporate during simple distillation. It remains in the flask as the residue.

Why the others are incorrect

A. Water vapour – Water vapour leaves the flask and enters the condenser.

B. Distilled water – This is collected in the receiving flask.

D. Steam – Steam is not left behind after the process.

Question 13

Correct Answer: B. Cool vapour into liquid

Explanation

The condenser removes heat from the vapour, causing it to condense back into a liquid called the distillate.

Why the others are incorrect

A. Heating is done by the burner.

C. The thermometer measures the boiling temperature.

D. The retort stand supports the apparatus.

Exam Tip: Condenser = **Cooling**.

Question 14

Correct Answer: C. Fractionating column

Explanation

The fractionating column provides many surfaces for repeated evaporation and condensation, producing a purer separation.

Why the others are incorrect

A. Thermometer – Measures temperature only.

B. Receiver – Collects the separated liquid.

D. Distillation flask – Holds the original mixture.

Question 15

Correct Answer: B. It boils at about 78°C.

Explanation

Ethanol has a boiling point of approximately **78°C**, which is lower than water's boiling point of **100°C**.

Why the others are incorrect

A. Water has the higher boiling point.

C. Freezing is not involved in fractional distillation.

D. Every pure liquid has a boiling point.

Exam Tip: Memorise these values:

- Ethanol = **78°C**
- Water = **100°C**

Question 16

Correct Answer: B. The highest point reached by the solvent

Explanation

The solvent front marks the furthest distance travelled by the solvent. It should be marked immediately when the paper is removed.

Why the others are incorrect

A. This is the baseline (origin).

C. The first coloured spot is one of the separated components.

D. The bottom of the paper is not the solvent front.

Question 17

Correct Answer: C. To prevent the sample from dissolving directly into the solvent

Explanation

If the sample spot is below the solvent level, it dissolves into the solvent reservoir instead of being carried up the paper, preventing proper separation.

Why the others are incorrect

A. Evaporation is not the reason.

B. Paper strength is not affected by the solvent level.

D. The solvent level does not change the boiling point.

Exam Tip: Baseline above solvent level is a common practical examination point.

Question 18

Correct Answer: D. Separating stones from sand

Explanation

Stones and sand are separated by **sieving**, not chromatography.

Why the others are incorrect

A. Food colourings are analysed by chromatography.

B. Plant pigments are separated by chromatography.

C. Chromatography is used to detect drugs in biological samples.

Question 19

Correct Answer: C. Capillary action

Explanation

Capillary action causes the solvent to move upwards through the tiny spaces in the filter paper fibres, carrying the dissolved substances with it.

Why the others are incorrect

A. Filtration separates insoluble solids from liquids.

B. Diffusion is the movement of particles from a region of high concentration to low concentration.

D. Sedimentation is the settling of solid particles under gravity.

Question 20

Correct Answer: B. Fractional distillation is used for liquids with similar boiling points.

Explanation

Fractional distillation separates **miscible liquids with close boiling points**, such as ethanol and water, using a fractionating column.

Why the others are incorrect

A. Simple distillation does **not** use a fractionating column.

C. Crude oil is separated by **fractional**, not simple, distillation.

D. Fractional distillation is widely used in industries, especially petroleum refineries, as well as laboratories.

Exam Tip: If a question mentions a **fractionating column**, the answer is almost always **fractional distillation**.

Quick Revision Box

Separation Method	Used For
Simple distillation	Solvent from a solution (e.g. water from salt solution)
Fractional distillation	Miscible liquids with close boiling points (e.g. ethanol and water)
Paper chromatography	Separating dyes, pigments and other dissolved substances
Filtration	Insoluble solids from liquids
Magnetic separation	Magnetic substances from non-magnetic substances
Separating funnel	Immiscible liquids such as oil and water

Question 21

A student wants to obtain **pure water** from muddy water that also contains dissolved salt. Which sequence of separation methods should be used?

A. Filtration followed by simple distillation

B. Simple distillation followed by filtration

C. Chromatography followed by filtration

D. Sieving followed by evaporation

Question 22

A mixture contains ethanol and water. During fractional distillation, the thermometer reads **78°C**. Which liquid is most likely being collected?

A. Water

B. Ethanol

C. Steam

D. Salt

Question 23

A student forgot to circulate cold water through the condenser during distillation. What is the most likely result?

A. More distillate will be collected.

B. Vapour will not condense efficiently.

C. Salt will evaporate.

D. The boiling point of water will decrease.

Question 24

Which statement best describes the function of the **fractionating column**?

A. It cools the vapour.

B. It measures temperature.

C. It allows repeated evaporation and condensation to improve separation.

D. It stores the distillate.

Question 25

During paper chromatography, three separate coloured spots are obtained from one black ink sample. This shows that the ink:

A. Is a pure substance.

B. Contains three different dyes.

C. Has evaporated completely.

D. Reacted with the pape

Question 26

A learner accidentally draws the baseline using a blue pen instead of a pencil. What is likely to happen?

A. Nothing unusual.

B. The blue ink may dissolve and interfere with the results.

C. The solvent will stop moving.

D. The paper will tear immediately.

Question 27

Which statement about the **mobile phase** is correct?

A. It remains fixed during chromatography.

B. It is the filter paper.

C. It is the solvent that moves through the paper.

D. It is the coloured spot after separation.

Question 28

Which laboratory technique would be most suitable for separating **green leaf pigments**?

A. Simple distillation

B. Filtration

C. Paper chromatography

D. Decantation

Question 29

In an oil refinery, why do different fractions condense at different heights in the fractionating tower?

A. They have different colours.

B. They have different boiling points.

C. They have different densities only.

D. They have different melting points.

Question 30

A student states:

"Simple distillation and fractional distillation use exactly the same apparatus."

Which response is correct?

A. The statement is completely correct.

B. The statement is incorrect because fractional distillation includes a fractionating column.

ANSWERS AND EXPLANATIONS

Question 21

Correct Answer: A. Filtration followed by simple distillation

Explanation

Muddy water contains **insoluble impurities (mud)** and **dissolved salt**.

- **Filtration** removes the mud.
- **Simple distillation** removes the dissolved salt and produces pure water.

Why the others are incorrect

B. Distillation before filtration is inefficient because the mud should be removed first.

C. Chromatography is not used to remove mud or dissolved salt.

D. Sieving cannot remove fine mud or dissolved salt.

Exam Tip: Always remove **insoluble solids first**, then remove dissolved substances.

Question 22

Correct Answer: B. Ethanol

Explanation

Ethanol boils at approximately **78°C**, while water boils at **100°C**.

When the thermometer reads around **78°C**, ethanol is the liquid being distilled.

Why the others are incorrect

A. Water boils later at 100°C.

C. Steam refers to water vapour.

D. Salt does not evaporate during fractional distillation.

Question 23

Correct Answer: B. Vapour will not condense efficiently.

Explanation

Without cooling water flowing through the condenser, the vapour remains as a gas and little or no liquid distillate is collected.

Why the others are incorrect

A. Less, not more, distillate is produced.

C. Salt does not evaporate under these conditions.

D. The boiling point of water remains unchanged.

Question 24

Correct Answer: C. It allows repeated evaporation and condensation to improve separation.

Explanation

The fractionating column contains glass beads or similar packing material that provides a large surface area. This allows repeated evaporation and condensation, producing a purer distillate.

Why the others are incorrect

A. Cooling is done by the condenser.

B. Temperature is measured by the thermometer.

D. The receiver collects the distillate.

Question 25

Correct Answer: B. Contains three different dyes.

Explanation

Each coloured spot represents a different dye. Three spots indicate that the black ink is a mixture of three coloured components.

Why the others are incorrect

A. A pure substance would normally produce only one spot.

C. Evaporation does not produce coloured spots.

D. No chemical reaction with the paper is responsible for the separation.

Exam Tip: One spot = Pure substance. More than one spot = Mixture.

Question 26

Correct Answer: B. The blue ink may dissolve and interfere with the results.

Explanation

Ink dissolves in the solvent and moves up the paper, making it difficult to identify the actual sample spots. Pencil graphite does not dissolve.

Why the others are incorrect

A. The experiment will be affected.

C. The solvent will continue moving.

D. The paper does not tear because of the pen.

Question 27

Correct Answer: C. It is the solvent that moves through the paper.

Explanation

The mobile phase is the moving solvent that carries the dissolved substances along the paper.

Why the others are incorrect

A. The stationary phase remains fixed.

B. Filter paper is the stationary phase.

D. The coloured spot is one of the separated components.

Question 28

Correct Answer: C. Paper chromatography

Explanation

Paper chromatography is ideal for separating pigments found in leaves, such as chlorophyll and carotene.

Why the others are incorrect

A. Distillation separates liquids.

B. Filtration removes insoluble solids.

D. Decantation separates a liquid from settled solids.

Question 29

Correct Answer: B. They have different boiling points.

Explanation

As vapours rise through the fractionating tower, the temperature decreases. Each fraction condenses where the temperature falls below its boiling point.

Why the others are incorrect

A. Colour has no effect.

C. Density alone does not determine where condensation occurs.

D. Melting point is not involved in distillation.

Question 30

Correct Answer: B. The statement is incorrect because fractional distillation includes a fractionating column.

Explanation

Both methods use similar equipment such as a flask, thermometer and condenser. However, **fractional distillation has an additional**

fractionating column, which is essential for separating liquids with close boiling points.

Why the others are incorrect

A. The apparatus is not exactly the same.

C. Neither method uses filter paper.

D. Both methods use a condenser.

Exam Tip: If you see a **fractionating column**, think **fractional distillation**.

Higher-Level Revision Notes

Remember these key facts:

Separation Method	Principle
Filtration	Particle size
Simple Distillation	Difference in boiling points
Fractional Distillation	Close boiling points with repeated evaporation and condensation
Paper Chromatography	Differences in solubility and attraction to the paper
Magnetic Separation	Magnetism

Common Boiling Points

Substance	Boiling Point (°C)
Ethanol	78
Water	100

Chromatography Facts

- **Stationary phase** = Filter paper

- **Mobile phase** = Solvent
- **Baseline** = Drawn with a pencil
- **Solvent front** = Highest point reached by the solvent
- **One spot** = Pure substance
- **Several spots** = Mixture

Question 31

A student heats a salt solution during simple distillation. At which stage does **condensation** occur?

A. When the water changes into vapour

B. When the vapour cools in the condenser to form liquid

C. When the salt remains in the flask

D. When the burner heats the flask

Question 32

Which one of the following pieces of apparatus is **NOT** used in simple distillation?

A. Liebig condenser

B. Thermometer

C. Fractionating column

D. Distillation flask

Question 33

During fractional distillation, why do water vapours return to the distillation flask?

A. Water has a higher boiling point than ethanol.

B. Water is insoluble in ethanol.

C. Water is heavier than ethanol.

D. Water evaporates more quickly.

Question 34

A learner observes **five different coloured spots** after performing paper chromatography on a food colouring. This means the food colouring:

A. Is a pure substance.

B. Contains five different coloured substances.

C. Has evaporated.

D. Has reacted with oxygen.

Question 35

Which of the following is the **mobile phase** during paper chromatography?

A. Filter paper

B. Solvent

C. Baseline

D. Pencil mark

Question 36

Which statement about the **solvent front** is correct?

A. It is the starting point of the sample.

B. It is the highest point reached by the solvent.

C. It is the lowest point of the chromatography paper.

D. It is the coloured spot nearest the top.

Question 37

Why should the chromatography paper **not touch the sides** of the beaker?

A. The solvent will boil.

B. The solvent will change colour.

C. Uneven movement of the solvent may affect the results.

D. The paper will dissolve immediately.

Question 38

Which fraction obtained from crude oil is mainly used as **fuel for aeroplanes**?

A. Petrol

B. Diesel

C. Kerosene

D. Bitumen

Question 39

Which separation method depends mainly on **capillary action**?

A. Filtration

B. Fractional distillation

C. Paper chromatography

D. Decantation

Question 40

A student wants to separate a mixture of **green leaf pigments**. Which statement is correct?

A. The green colour comes from only one pigment.

B. Paper chromatography can separate the different pigments.

C. Distillation is the best method.

D. Filtration separates the pigments into different colours.

ANSWERS AND EXPLANATIONS

Question 31

Correct Answer: B. When the vapour cools in the condenser to form liquid

Explanation

Condensation is the process in which a gas changes back into a liquid. During distillation, water vapour cools inside the condenser and becomes liquid water.

Why the others are incorrect

A. This is **evaporation**, not condensation.

C. Salt remaining in the flask is not condensation.

D. Heating causes evaporation, not condensation.

Exam Tip:
Evaporation: Liquid → Gas
Condensation: Gas → Liquid

Question 32

Correct Answer: C. Fractionating column

Explanation

Simple distillation does **not** use a fractionating column. It consists mainly of a distillation flask, thermometer, condenser and receiving flask.

Why the others are incorrect

A. The condenser is essential.

B. The thermometer measures vapour temperature.

D. The distillation flask holds the mixture.

Question 33

Correct Answer: A. Water has a higher boiling point than ethanol.

Explanation

Water condenses inside the fractionating column because it requires a higher temperature to remain as vapour. Ethanol, with its lower boiling point, continues upwards to the condenser.

Why the others are incorrect

B. Ethanol and water are completely miscible.

C. Density does not determine the separation.

D. Water evaporates less easily than ethanol.

Question 34

Correct Answer: B. Contains five different coloured substances.

Explanation

Each coloured spot on a chromatogram represents a different component of the mixture. Five spots indicate five different dyes or pigments.

Why the others are incorrect

A. A pure substance would normally give one spot.

C. Evaporation alone does not produce separate spots.

D. Oxidation is not responsible for the separation.

Exam Tip:
More spots = More components in the mixture.

Question 35

Correct Answer: B. Solvent

Explanation

The solvent moves up the paper carrying the dissolved substances. Therefore, it is called the **mobile phase**.

Why the others are incorrect

A. Filter paper is the stationary phase.

C. The baseline is simply the starting line.

D. The pencil mark remains stationary.

Question 36

Correct Answer: B. It is the highest point reached by the solvent.

Explanation

The solvent front shows the maximum distance travelled by the solvent and should be marked immediately after removing the paper from the beaker.

Why the others are incorrect

A. This describes the baseline.

C. The solvent front is not at the bottom.

D. It is not a coloured spot.

Question 37

Correct Answer: C. Uneven movement of the solvent may affect the results.

Explanation

If the paper touches the sides of the beaker, the solvent may rise unevenly, causing distorted or inaccurate separation.

Why the others are incorrect

A. The solvent does not boil.

B. The solvent does not change colour because the paper touches the beaker.

D. Filter paper does not dissolve under normal chromatography conditions

Question 38

Correct Answer: C. Kerosene

Explanation

Kerosene, also called aviation fuel, is one of the fractions obtained during the fractional distillation of crude oil and is widely used to fuel aircraft.

Why the others are incorrect

A. Petrol is mainly used in cars.

B. Diesel is used in buses, trucks and some generators.

D. Bitumen is used for road surfacing and roofing.

Exam Tip:
Petrol → Cars
Kerosene → Aircraft
Diesel → Heavy vehicles
Bitumen → Roads

Question 39

Correct Answer: C. Paper chromatography

Explanation

Paper chromatography depends on **capillary action**, which causes the solvent to rise through the filter paper, carrying the dissolved substances.

Why the others are incorrect

A. Filtration depends on particle size.

B. Fractional distillation depends on differences in boiling points.

D. Decantation depends on gravity.

Question 40

Correct Answer: B. Paper chromatography can separate the different pigments.

Explanation

Leaves contain several pigments, such as chlorophyll a, chlorophyll b, xanthophyll and carotene. Paper chromatography separates these pigments into individual coloured spots.

Why the others are incorrect

A. Green leaves contain several pigments, not just one.

C. Distillation separates liquids, not pigments.

D. Filtration cannot separate dissolved pigments.

ZIMSEC Examination Challenge

Question 40 Extension

A learner performs paper chromatography on two different black ink samples.

After the experiment:

- Sample **A** produces **one spot**.
- Sample **B** produces **four spots**.

Which conclusion is correct?

A. Both samples are pure substances.

B. Sample A is a mixture, while Sample B is pure.

C. Sample A is likely a pure substance, while Sample B is a mixture.

D. Both samples are mixtures.

Answer: C

Explanation: A pure substance generally produces **one spot**, whereas a mixture separates into **two or more spots** because it contains multiple components.

Practical Skills Checklist

By the end of this topic, learners should be able to:

✅ Draw and label a simple distillation apparatus.

✅ Draw and label a fractional distillation apparatus.

✅ Draw and label a paper chromatography setup.

- ✅ State the function of every piece of apparatus.
- ✅ Select the correct separation method for a given mixture.
- ✅ Explain the principles behind each separation technique.

Question 41

A student wants to obtain **pure ethanol** from a mixture of ethanol and water. Which method should be used?

A. Filtration

B. Fractional distillation

C. Paper chromatography

D. Evaporation

Question 42

During paper chromatography, one dye travelled **8 cm**, while another travelled **3 cm**. The dye that travelled 8 cm:

A. Was more strongly attracted to the paper.

B. Was more soluble in the solvent.

C. Had a higher boiling point.

D. Was heavier.

Question 43

A learner is separating crude oil into useful products. Which property makes this separation possible?

A. Differences in density

B. Differences in boiling points

C. Differences in colour

D. Differences in melting points

Question 44

Which one of the following laboratory techniques produces the **purest water**?

A. Filtration

B. Decantation

C. Simple distillation

D. Sieving

Question 45

Which statement about paper chromatography is **correct**?

A. It separates substances according to differences in boiling points.

B. It separates substances according to differences in solubility and attraction to the paper.

C. It separates magnetic substances.

D. It separates insoluble solids from liquids.

Question 46

Why is the chromatography container usually covered during the experiment?

A. To increase the boiling point of the solvent.

B. To reduce evaporation of the solvent and improve results.

C. To stop the paper from falling.

D. To make the solvent change colour.

Question 47

A student separates a black marker ink using paper chromatography and obtains **only one spot**. This suggests that the marker ink is:

A. A mixture of many dyes.

B. A pure substance under the conditions of the experiment.

C. Water.

D. Colourless.

Question 48

Which statement about the **fractionating column** is correct?

A. It decreases the purity of the distillate.

B. It increases the efficiency of separation by repeated condensation and evaporation.

C. It heats the condenser.

D. It stores the condensed liquid.

Question 49

Which of the following pairs is **correctly matched**?

	Method	Example
A	Filtration	Ethanol and water
B	Paper chromatography	Plant pigments
C	Simple distillation	Sand and water
D	Fractional distillation	Iron filings and sulphur

Question 50

A student makes the following statements:

1. Distillation depends on differences in boiling points.
2. Fractional distillation uses a fractionating column.
3. Paper chromatography uses filter paper as the stationary phase.

Which statements are correct?

A. 1 only

B. 1 and 2 only

C. 2 and 3 only

D. 1, 2 and 3

ANSWERS AND EXPLANATIONS

Question 41

Correct Answer: B. Fractional distillation

Explanation

Ethanol and water are **miscible liquids** with close boiling points (78°C and 100°C). Fractional distillation is the appropriate method because the fractionating column improves separation.

Why the others are incorrect

A. Filtration – Removes insoluble solids only.

C. Paper chromatography – Separates dissolved substances such as dyes.

D. Evaporation – Cannot produce pure ethanol.

Question 42

Correct Answer: B. Was more soluble in the solvent.

Explanation

A substance that travels further is generally **more soluble in the solvent** and/or **less strongly attracted to the paper**.

Why the others are incorrect

A. A substance strongly attracted to the paper moves a shorter distance.

C. Boiling point is not involved in chromatography.

D. Mass does not determine the distance travelled.

Exam Tip: The farther a spot travels, the greater its attraction to the **solvent**.

Question 43

Correct Answer: B. Differences in boiling points

Explanation

Crude oil contains many hydrocarbons with different boiling points. Each fraction condenses at a different temperature in the fractionating column.

Why the others are incorrect

A. Density is not the main principle.

C. Colour is irrelevant.

D. Melting point is not used in distillation.

Question 44

Correct Answer: C. Simple distillation

Explanation

Simple distillation removes dissolved substances and produces pure water by evaporation and condensation.

Why the others are incorrect

A. Filtration removes only insoluble particles.

B. Decantation separates liquids from settled solids.

D. Sieving separates solids of different sizes.

Question 45

Correct Answer: B. It separates substances according to differences in solubility and attraction to the paper.

Explanation

Paper chromatography separates substances because different components dissolve differently in the solvent and interact differently with the paper.

Why the others are incorrect

A. Distillation depends on boiling points.

C. Magnetic separation uses magnetism.

D. Filtration separates insoluble solids from liquids.

Question 46

Correct Answer: B. To reduce evaporation of the solvent and improve results.

Explanation

Covering the container helps maintain a solvent-saturated atmosphere, reducing evaporation and giving more reliable separation.

Why the others are incorrect

A. The boiling point is unaffected.

C. The cover is not mainly for supporting the paper.

D. The solvent does not change colour.

Question 47

Correct Answer: B. A pure substance under the conditions of the experiment.

Explanation

A single spot usually indicates that only one component has been detected in the sample under the chromatography conditions used.

Why the others are incorrect

A. A mixture would usually produce two or more spots.

C. A coloured marker is not water.

D. The marker clearly contains a coloured substance.

Exam Tip: In school chromatography:

- **One spot → Pure substance**
- **Several spots → Mixture**

Question 48

Correct Answer: B. It increases the efficiency of separation by repeated condensation and evaporation.

Explanation

The fractionating column improves separation because vapours repeatedly condense and re-evaporate, enriching the vapour in the more volatile component.

Why the others are incorrect

A. It increases, not decreases, purity.

C. It does not heat the condenser.

D. The receiving flask stores the condensed liquid.

Question 49

Correct Answer: B. Paper chromatography — Plant pigments

Explanation

Paper chromatography is commonly used to separate plant pigments such as chlorophyll and carotene.

Why the others are incorrect

A. Ethanol and water require fractional distillation.

C. Sand and water are separated by filtration.

D. Iron filings and sulphur are separated with a magnet.

Question 50

Correct Answer: D. 1, 2 and 3

Explanation

All three statements are correct:

- **Statement 1:** Distillation separates substances using differences in boiling points.
- **Statement 2:** Fractional distillation uses a fractionating column.
- **Statement 3:** In paper chromatography, the filter paper acts as the stationary phase.

Why the others are incorrect

A. Ignores two correct statements.

B. Omits Statement 3, which is correct.

C. Omits Statement 1, which is also correct.

Final Revision Summary

Key Separation Methods

Method	Principle	Example
Filtration	Difference in particle size	Sand and water
Simple Distillation	Difference in boiling points	Salt and water
Fractional Distillation	Close boiling points	Ethanol and water
Paper Chromatography	Differences in solubility and attraction to paper	Dyes and plant pigments
Magnetic Separation	Magnetism	Iron and sulphur
Separating Funnel	Difference in density of immiscible liquids	Oil and water

Important Laboratory Apparatus

Apparatus	Function
Distillation flask	Holds the mixture to be heated
Thermometer	Measures vapour temperature
Fractionating column	Improves separation by repeated evaporation and condensation
Liebig condenser	Cools vapour into liquid
Receiving flask	Collects the distillate
Filter paper	Stationary phase in chromatography
Solvent	Mobile phase in chromatography

Common Examination Facts to Remember

- Water boils at **100°C**.
- Ethanol boils at **78°C**.
- Cold water enters the condenser from the **bottom** and leaves from the **top**.

- Draw the chromatography **baseline with a pencil**, not a pen.
- The sample spot must be **above the solvent level**.
- The **solvent front** is marked immediately after removing the paper.
- A **single spot** on a chromatogram usually indicates a pure substance, while **multiple spots** indicate a mixture.

STRUCTURED QUESTIONS

PART 1 (Questions 1–5)

Total: 20 Marks

Instructions

Answer all questions.

Show all working where necessary.

Question 1 (2 Marks)

(a) Define **simple distillation**. (1)

(b) State **one use** of simple distillation. (1)

Marking Guide

(a) (1 mark)

Award **1 mark** for either of the following:

- Simple distillation is a method used to separate a solvent from a solution using differences in boiling points.

OR

- Simple distillation is a method used to obtain a pure liquid by heating and condensing its vapour.

(b) (1 mark)

Any ONE:

- Obtaining pure water from salt solution
- Purifying water
- Producing distilled water

- Separating water from dissolved salts

(1)

Examiner's Comment

Many learners write **"separating solids from liquids."**

This is **too vague**.

Always mention **boiling point** or **obtaining a pure liquid**.

Question 2 (3 Marks)

A learner sets up the apparatus shown below to separate salt from water.

Answer the questions that follow.

(a) Name the separation method used. (1)

(b) State the function of the condenser. (1)

(c) Explain why salt remains in the distillation flask. (1)

Marking Guide

(a)

Simple distillation

(1)

(b)

Any one

- Cools vapour
- Condenses vapour into liquid
- Changes vapour into liquid water

(1)

(c)

Award 1 mark for:

Salt has a much higher boiling point than water and therefore does not evaporate.

OR

Salt is non-volatile under these conditions and remains in the flask.

Examiner's Comment

Many learners incorrectly write

"Salt melts."

Salt neither melts nor evaporates during this experiment.

Question 3 (4 Marks)

Study the statements below.

Liquid	**Boiling Point**
Ethanol	78°C
Water	100°C

(a) Which liquid will distil first? (1)

(b) Give a reason for your answer. (1)

(c) Name the method used to separate ethanol from water. (1)

(d) Name the part of the apparatus that makes this separation more efficient. (1

Marking Guide

(a)

Ethanol

(1)

(b)

It has the lower boiling point.

(1)

(c)

Fractional distillation

(1)

(d)

Fractionating column

(1)

Examiner's Comment

Remember

Simple distillation cannot efficiently separate liquids whose boiling points are close together.

Question 4 (5 Marks)

A learner carries out paper chromatography using black ink.

After some time, four coloured spots appear.

Answer the questions.

(a) What does the appearance of four spots show about the black ink? (1)

(b) Name the stationary phase. (1)

(c) Name the mobile phase. (1)

(d) Why is the baseline drawn using a pencil? (1)

(e) Why should the sample spot remain above the solvent level? (1)

Marking Guide

(a)

The black ink is a mixture of four dyes.

(1)

(b)

Filter paper

(1)

(c)

The solvent

(1)

(d)

Graphite does not dissolve in the solvent.

OR

Pencil does not interfere with the results.

(1)

(e)

To prevent the sample dissolving directly into the solvent.

OR

To allow the solvent to carry the sample up the paper.

(1)

Examiner's Comment

One of the most common ZIMSEC mistakes is learners drawing the baseline with a pen.

Always use a **pencil**.

Question 5 (6 Marks)

A science class wants to investigate how paper chromatography separates different food colourings.

Answer the following questions.

(a) List **four pieces of apparatus** needed for the experiment. (4)

(b) State **two precautions** taken during the experiment. (2)

Marking Guide

(a)

Any FOUR

- Filter paper
- Beaker
- Pencil
- Ruler
- Solvent
- Capillary tube
- Food colouring
- Watch glass
- Glass rod

Any four ×1

(4)

(b)

Any TWO

- Draw the baseline with pencil.
- Ensure the sample spot is above the solvent.
- Cover the beaker.
- Do not allow the paper to touch the sides.
- Mark the solvent front immediately.
- Use only a small sample spot.

Any two ×1

(2)

Examiner's Comments

Practical questions usually test whether learners understand **how experiments are carried out**, not just the theory.

Learners should practise identifying:

- Correct apparatus.
- Correct procedures.
- Safety precautions.
- Reasons for each step.

Total: 20 Marks

Question 6 (3 Marks)

A learner is given the following mixtures:

- Salt and water
- Ethanol and water
- Sand and water

For each mixture, state the **most suitable separation method**.

Mixture	Separation Method
(a) Salt and water	....
(b) Ethanol and water	....
(c) Sand and water	....

Marking Guide

(a) **Simple distillation** (1)

(b) **Fractional distillation** (1)

(c) **Filtration** (1)

Total: 3 Marks

Examiner's Comment

Learners often confuse **simple** and **fractional distillation**.

Remember:

- **Simple distillation** → solvent from a solution.
- **Fractional distillation** → two miscible liquids with close boiling points.

Question 7 (4 Marks)

Study the table below.

Substance	Boiling Point (°C)
Ethanol	78
Water	100

Answer the questions.

(a) Which substance boils first? (1)

(b) Explain why it boils first. (1)

(c) Name the separation method used. (1)

(d) State one use of this separation method. (1)

Marking Guide

(a) Ethanol (1)

(b) It has a lower boiling point than water. (1)

(c) Fractional distillation. (1)

(d) Any ONE:

- Separating ethanol from water.

- Refining crude oil into useful fractions.
- Separating miscible liquids with different boiling points.

(1)

Total: 4 Marks

Examiner's Comment

When explaining **why** ethanol boils first, always mention its **lower boiling point**.

Do **not** say it is lighter or less dense.

Question 8 (5 Marks)

A learner carries out paper chromatography to separate black ink.

After the experiment, four coloured spots appear.

Answer the following questions.

(a) What conclusion can be made about the black ink? (1)

(b) Name the stationary phase. (1)

(c) Name the mobile phase. (1)

(d) Explain why different dyes move different distances. (2)

Marking Guide

(a) Black ink is a mixture of four dyes. (1)

(b) Filter paper. (1)

(c) Solvent. (1)

(d) Award **2 marks** for any two correct points:

- Different dyes have different solubilities in the solvent.
- Different dyes have different attractions to the filter paper.
- Some dyes move faster than others because they dissolve more readily in the solvent.

(Any two ×1)

Total: 5 Marks

Examiner's Comment

Many learners simply write:

"They move at different speeds."

This is incomplete.

Always explain **why**:

- Different **solubilities**.
- Different **attractions to the paper**.

Question 9 (6 Marks)

A student sets up a simple distillation apparatus to obtain pure water from seawater.

Answer the questions.

(a) State the function of each of the following:

(i) Thermometer (1)

(ii) Condenser (1)

(iii) Distillation flask (1)

(b) Explain why cold water enters the condenser from the bottom. (2)

(c) Name the liquid collected in the receiving flask. (1)

Marking Guide

(a)

(i)

Measures the temperature of the vapour.

(1)

(ii)

Cools the vapour and changes it into liquid.

(1)

(iii)

Holds the seawater while it is heated.

(1)

(b)

Award **2 marks** for:

- Cold water enters at the bottom so the cooling jacket fills completely. (1)
- This ensures efficient cooling and condensation of the vapour. (1)

(c)

Distilled water (or pure water).

(1)

Total: 6 Marks

Examiner's Comment

This is a favourite practical question.

Learners should know the function of **every piece of apparatus** used in distillation.

Question 10 (7 Marks)

A petroleum refinery separates crude oil into useful products using fractional distillation.

Answer the following questions.

(a) Define **fractional distillation**. (2)

(b) State **three fractions** obtained from crude oil. (3)

(c) Give one use for **each** fraction named in (b). (3)

Marking Guide

(a) (2 Marks)

Award:

- **1 mark**: It is a method used to separate miscible liquids.
- **1 mark**: Separation occurs because the liquids have different boiling points.

(b) (Any three ×1)

Examples:

- Refinery gas
- Petrol (gasoline)
- Naphtha
- Kerosene (paraffin)
- Diesel
- Fuel oil
- Lubricating oil
- Bitumen

(3)

(c) (Any corresponding uses ×1)

Fraction	Use
Refinery gas	Cooking/heating
Petrol	Fuel for cars
Kerosene	Jet fuel, lamps
Diesel	Fuel for buses, trucks, tractors
Fuel oil	Ships, industrial boilers
Lubricating oil	Lubricating engines and machinery
Bitumen	Road surfacing, roofing

(3)

Maximum: 3 Marks

Total: 8 Marks

Practical Skills Focus

By the end of this section, learners should be able to:

✅ Choose the correct separation method for a given mixture.

✅ Explain why a particular method is suitable.

✅ Describe the function of distillation apparatus.

✅ Interpret chromatography results.

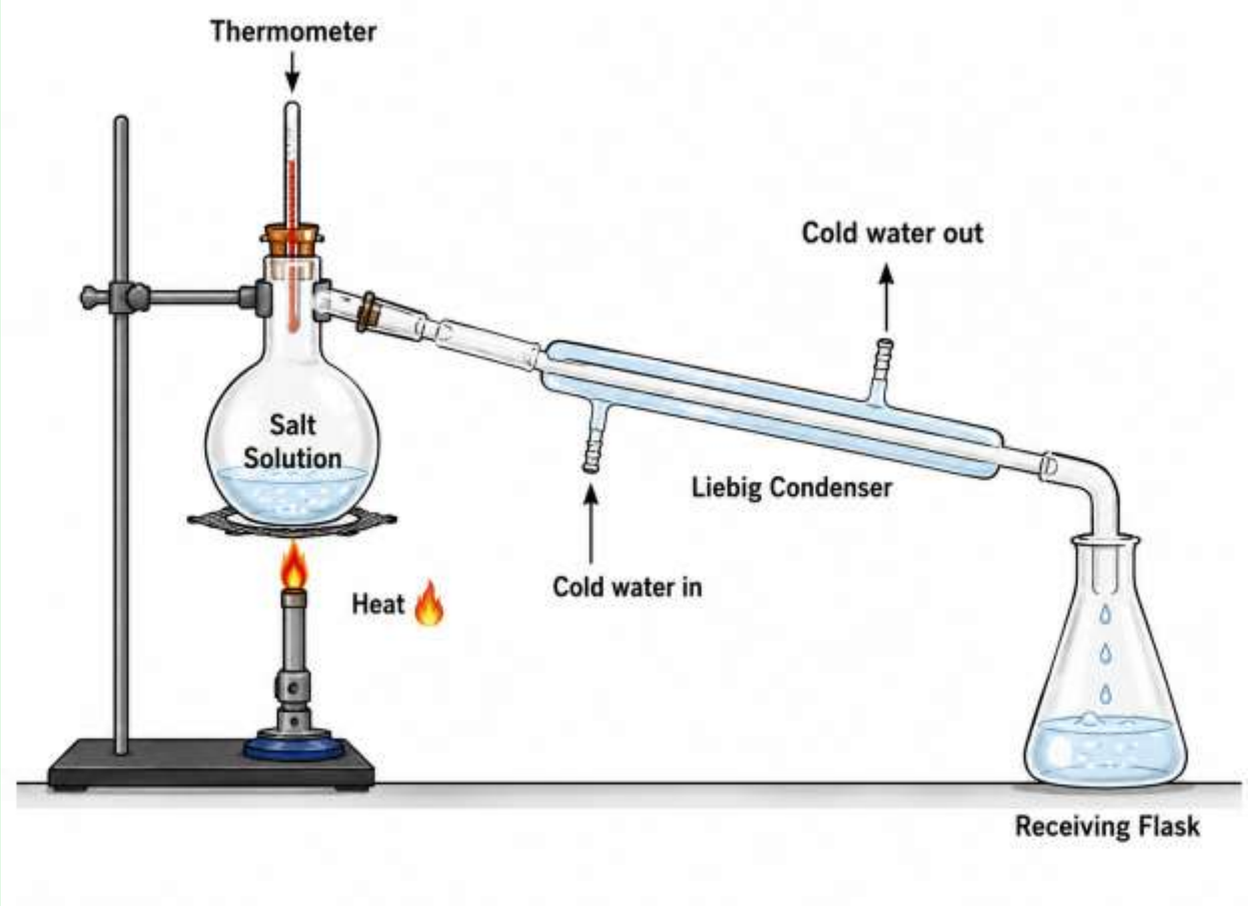
✅ Identify products obtained from crude oil and their uses.

Common ZIMSEC Mistakes

1. Confusing **simple** and **fractional distillation**.
2. Forgetting that **paper is the stationary phase**.
3. Stating that **density** is the reason ethanol distils first.
4. Drawing the chromatography baseline with a pen.
5. Forgetting that **cold water enters the condenser from the bottom**.
6. Naming crude oil fractions without giving their uses.

Question 11 (4 Marks)

The diagram below shows a simple distillation apparatus.



Answer the following questions.

(a) Name the apparatus labelled **A** (the long horizontal tube). (1)

(b) State its function. (1)

(c) Why does cold water enter from the bottom of the condenser? (1)

(d) Name the liquid collected in the receiving flask. (1)

Marking Guide

(a)

Liebig condenser / Condenser (1)

(b)

Any ONE

- Cools vapour
- Condenses vapour into liquid
- Changes vapour into liquid

(1)

(c)

Cold water enters from the bottom so that the cooling jacket fills completely, ensuring efficient cooling.

(1)

(d)

Distilled water / Pure water

(1)

Total: 4 Marks

Examiner's Comment

Many learners incorrectly write that the condenser "stores water." Its function is to **cool and condense vapour**.

Question 12 (5 Marks)

A student separates ethanol and water using fractional distillation.

(a) Explain why **fractional distillation** is used instead of simple distillation. (2)

(b) State the function of the **fractionating column**. (2)

(c) Which liquid is collected first? (1)

Marking Guide

(a)

Award **2 marks**

- Ethanol and water are **miscible liquids**. (1)
- They have **close boiling points**, so fractional distillation gives better separation. (1)

(b)

Award **2 marks**

- Provides a large surface area. (1)
- Allows repeated evaporation and condensation, improving separation. (1)

(c)

Ethanol

(1)

Total: 5 Marks

Examiner's Comment

Learners should always mention **both**:

- Miscible liquids
- Close boiling points

These are the key reasons for using fractional distillation.

Question 13 (6 Marks)

A learner performed paper chromatography on a green leaf extract.

The chromatogram produced the following results:

Pigment	Distance moved by pigment (cm)
A	2.0
B	4.5
C	6.0
Solvent front	8.0

Answer the following questions.

(a) Which pigment travelled the greatest distance? (1)

(b) Which pigment was most strongly attracted to the paper? Explain your answer. (2)

(c) State two conclusions that can be made about the leaf extract. (2)

(d) Name the separation technique used. (1)

Marking Guide

(a)

Pigment **C**

(1)

(b)

Pigment **A** (1)

Reason:

It travelled the shortest distance because it was more strongly attracted to the paper (stationary phase).

(1)

(c)

Any TWO

- The leaf contains several pigments.
- The pigments have different solubilities.
- The pigments have different attractions to the paper.
- The leaf extract is a mixture.

(Any two ×1)

(d)

Paper chromatography

(1)

Total: 6 Marks

Examiner's Comment

A common mistake is assuming the pigment that travels furthest is most attracted to the paper. It is actually **less attracted to the paper and more attracted to the solvent**.

Question 14 (4 Marks)

Choose the **most suitable separation method** for each mixture and give one reason for your choice.

Method	Reason
(a) Oil and water	
(b) Iron filings and sulphur	

Marking Guide

(a)

Method:

Separating funnel

(1)

Reason:

Oil and water are **immiscible liquids** with different densities.

(1)

(b)

Method:

Magnetic separation

(1)

Reason:

Iron is magnetic but sulphur is not.

(1)

Total: 4 Marks

Examiner's Comment

Always explain **why** a method is suitable. Naming the method alone is not enough.

Question 15 (6 Marks)

A Form 4 learner was asked to compare **simple distillation** and **paper chromatography**.

Complete the table below.

Feature	Simple Distillation	Paper Chromatography
(a) Principle used		
(b) One application		
(c) Main apparatus		

Marking Guide

(a)

Simple distillation:

Difference in boiling points

(1)

Paper chromatography:

Difference in solubility and attraction to the paper

(1)

(b)

Simple distillation:

Any ONE

- Purifying water
- Obtaining distilled water
- Separating water from salt solution

(1)

Paper chromatography:

Any ONE

- Separating dyes
- Separating plant pigments
- Identifying food colourings
- Drug testing

(1)

(c)

Simple distillation:

Any ONE

- Distillation flask
- Condenser
- Thermometer

(1)

Paper chromatography:

Any ONE

- Filter paper
- Beaker
- Solvent

(1)

Total: 6 Marks

Practical Skills Challenge

A learner accidentally:

- Drew the baseline using a blue pen.
- Allowed the solvent level to rise above the sample spot.
- Forgot to mark the solvent front.
- Allowed the chromatography paper to touch the side of the beaker.

Question

State **four mistakes** made by the learner and explain how each would affect the results.

Common ZIMSEC Examination Mistakes

❌ Saying ethanol is collected first because it is lighter.

✓ Correct: Ethanol is collected first because it has a **lower boiling point**.

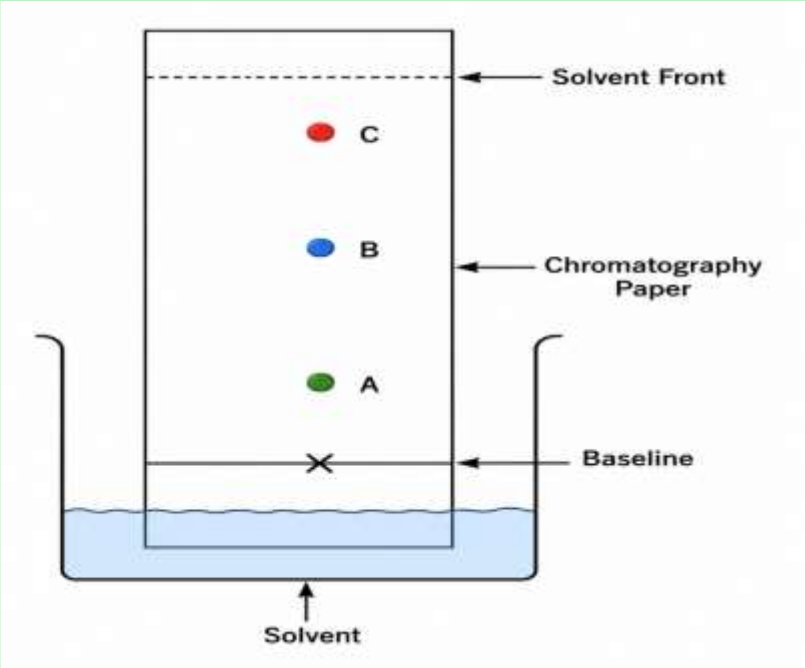
❌ Saying chromatography separates substances by colour.

✓ Correct: It separates substances because of differences in **solubility** and **attraction to the stationary phase**.

❌ Saying the condenser heats vapour.

✓ Correct: The condenser **cools vapour**, causing it to condense into a liquid.

Question 16 (5 Marks)



The diagram below shows a chromatography experiment

Study the chromatogram and answer the questions.

(a) What is the name of this separation technique? (1)

(b) Which spot travelled the furthest? (1)

(c) Which spot was most strongly attracted to the filter paper? Explain your answer. (2)

(d) What is the name of the line marked at the top of the paper? (1)

Marking Guide

(a)

Paper chromatography

(1)

(b)

Spot **C**

(1)

(c)

Spot **A** (1)

Reason:

It travelled the shortest distance because it was more strongly attracted to the stationary phase (filter paper) and less soluble in the solvent.

(1)

(d)

Solvent front

(1)

Total: 5 Marks

Examiner's Comment

Learners often confuse the **baseline** with the **solvent front**. The baseline is drawn before the experiment begins, while the solvent front is marked immediately after the experiment ends.

Question 17 (4 Marks)

A student wants to obtain **pure water** from seawater.

(a) Name the most suitable separation method. (1)

(b) Explain why this method is suitable. (2)

(c) Name the substance left behind in the distillation flask. (1)

Marking Guide

(a)

Simple distillation

(1)

(b)

Award **2 marks**

- Water has a lower boiling point than dissolved salts. (1)
- Water evaporates and condenses while the salts remain in the flask. (1)

(c)

Salt (or dissolved salts)

(1)

Total: 4 Marks

Examiner's Comment

Do not write **"the salt evaporates."** Salt remains in the flask because it has a much higher boiling point.

Question 18 (6 Marks)

A mixture contains **ethanol and water**.

The boiling points are shown below.

Liquid	Boiling Point (°C)
Ethanol	78
Water	100

Answer the following questions.

(a) Which liquid will be collected first? (1)

(b) Explain your answer. (2)

(c) Name the separation method used. (1)

(d) Explain the role of the fractionating column. (2)

Marking Guide

(a)

Ethanol

(1)

(b)

Award **2 marks**

- Ethanol has a lower boiling point than water. (1)
- Therefore, ethanol evaporates first and reaches the condenser before water. (1)

(c)

Fractional distillation

(1)

(d)

Award **2 marks**

- Provides a large surface area for repeated evaporation and condensation. (1)
- Improves the separation of liquids with close boiling points. (1)

Total: 6 Marks

Examiner's Comment

For full marks, learners must explain **both** why ethanol evaporates first **and** how the fractionating column improves the separation.

Question 19 (5 Marks)

A learner carried out paper chromatography correctly.

The following observations were made:

- Four coloured spots appeared.
- The solvent reached the solvent front.
- The baseline remained visible.

Answer the following questions.

(a) What do the four coloured spots indicate? (1)

(b) Why is it important to use a pencil when drawing the baseline? (1)

(c) State one reason why the solvent level must remain below the baseline. (1)

(d) State two applications of paper chromatography. (2)

Marking Guide

(a)

The sample is a mixture containing four different substances (or dyes).

(1)

(b)

Graphite does not dissolve in the solvent, so it does not interfere with the results.

(1)

(c)

To prevent the sample from dissolving directly into the solvent.

(1)

(d)

Any TWO

- Separating food colourings
- Separating plant pigments
- Drug testing

- Forensic investigations
- Identifying inks
- Quality control in industries

(Any two ×1)

Total: 5 Marks

Examiner's Comment

Applications of chromatography are commonly tested in ZIMSEC examinations. Learners should know several real-life uses.

Question 20 (5 Marks)

A student is given the following mixtures.

Mixture	Separation Method Used
Sand and water	Filtration
Salt and water	Simple distillation
Ethanol and water	Fractional distillation
Plant pigments	Paper chromatography

Answer the following questions.

(a) Explain why **filtration** is suitable for separating sand and water. (1)

(b) Explain why **simple distillation** is suitable for salt solution. (1)

(c) Explain why **fractional distillation** is used for ethanol and water. (1)

(d) Explain why **paper chromatography** is suitable for plant pigments. (2)

Marking Guide

(a)

Sand is insoluble and its particles are too large to pass through filter paper.

(1)

(b)

Water evaporates and condenses, leaving the dissolved salt behind because of the difference in boiling points.

(1)

(c)

Ethanol and water are miscible liquids with close boiling points.

(1)

(d)

Award **2 marks**

- Plant pigments have different solubilities in the solvent. (1)
- They have different attractions to the filter paper, causing them to move different distances. (1)

Total: 5 Marks

Practical Investigation Question (Teacher Extension)

A learner performed paper chromatography but obtained **only one coloured spot** instead of the expected four.

Suggest **four possible reasons** for this result.

Expected Answers

Any four:

- The ink may have been a pure substance.
- The wrong solvent was used.
- The solvent did not travel far enough.
- The sample spot was too large.
- The sample was placed below the solvent level.
- The chromatography paper touched the side of the beaker.
- The experiment was stopped too early.
- The solvent evaporated because the container was not covered.

Common Examiner's Tips

✔ Always compare boiling points when explaining distillation.

✔ Mention **solubility** and **attraction to the stationary phase** when explaining chromatography.

✔ Support every answer with a scientific reason—many ZIMSEC questions award marks for explanations, not just correct terms.

✔ Learn the functions of every apparatus used in practical work.

Question 21 (5 Marks)

A learner wanted to obtain pure water from seawater using the apparatus shown below.

During the experiment the following observations were made:

- Water boiled after some time.
- Water droplets formed inside the condenser.
- A colourless liquid collected in the receiving flask.
- White crystals remained in the distillation flask.

(a)

Name the process responsible for changing

(i) Water into vapour (1)

(ii) Vapour into liquid (1)

(b)

Explain why white crystals remained in the distillation flask. (2)

(c)

Name the colourless liquid collected. (1)

MARKING GUIDE

(a)

(i)

Evaporation / Boiling

(1)

(ii)

Condensation

(1)

(b)

Award 2 marks

- Salt has a much higher boiling point than water. (1)
- Therefore it does not evaporate and remains in the flask. (1)

(c)

Distilled water / Pure water

(1)

Total: 5 Marks

Examiner's Comment

Many candidates confuse **evaporation** with **condensation**.

Remember

Liquid → Gas = Evaporation

Gas → Liquid = Condensation

Question 22 (6 Marks)

A chemistry teacher asked learners to compare **simple distillation** and **fractional distillation**.

Complete the table below.

Feature	Simple Distillation	Fractional Distillation
(a) Liquids separated		
(b) Main apparatus		
(c) Industrial application		

MARKING GUIDE

(a)

Simple

Separates a solvent from a solution.

(1)

Fractional

Separates miscible liquids with close boiling points.

(1)

(b)

Simple

Distillation flask, thermometer, condenser.

(1)

Fractional

Distillation flask, thermometer, condenser and fractionating column.

(1)

(c)

Simple

Producing distilled water.

(1)

Fractional

Refining crude oil.

(1)

Total: 6 Marks

Examiner's Comment

Candidates should compare the methods rather than describing only one

Question 23 (5 Marks)

A learner carried out paper chromatography.

The chromatogram below was obtained.

Spot	Distance travelled by spot (cm)
A	2
B	5
C	7
Solvent front	8

(a)

Which spot travelled furthest? (1)

(b)

Which spot had the greatest attraction to the solvent? Explain. (2)

(c)

What conclusion can be made about the original sample? (2)

MARKING GUIDE

(a)

Spot **C**

(1)

(b)

Spot **C** (1)

Reason

It travelled the greatest distance because it was the most soluble in the solvent (or least attracted to the paper).

(1)

(c)

Award 2 marks

- The sample was a mixture. (1)
- It contained three different substances/pigments/dyes. (1)

Total: 5 Marks

Examiner's Comment

A learner who simply writes

"It moved faster."

will not obtain full marks.

Always explain **why**.

Question 24 (4 Marks)

A learner made the following mistakes during paper chromatography.

- Used a blue pen instead of a pencil.
- Allowed the solvent level to cover the sample.
- Did not cover the beaker.
- Allowed the paper to touch the side of the beaker.

Explain how **each mistake** affects the experiment.

MARKING GUIDE

Award one mark for each explanation.

(a)

Pen ink dissolves and interferes with the chromatogram.

(1)

(b)

Sample dissolves into the solvent instead of travelling up the paper.

(1)

(c)

The solvent evaporates quickly, giving poor separation.

(1)

(d)

The solvent rises unevenly, producing inaccurate results.

(1)

Total: 4 Marks

Examiner's Comment

This type of question tests practical skills and scientific reasoning, not memory.

Question 25 (5 Marks)

A laboratory contains the following mixtures.

Mixture
Salt and water
Ethanol and water
Sand and water
Plant pigments
Iron and sulphur

Complete the table below.

Mixture Separation Method Scientific Principle

MARKING GUIDE

Mixture	Method	Principle
Salt + water	Simple distillation	Difference in boiling points
Ethanol + water	Fractional distillation	Difference in boiling points of miscible liquids
Sand + water	Filtration	Difference in particle size
Plant pigments	Paper chromatography	Difference in solubility and attraction to paper
Iron + sulphur	Magnetic separation	Magnetism